

# A Switch from High-Fidelity to Error-Prone DNA Double-Strand Break Repair Underlies Stress-Induced Mutation

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## Summary

Special mechanisms of mutation are induced in microbes under growth-limiting stress causing genetic instability, including occasional adaptive mutations that may speed evolution. Both the mutation mechanisms and their control by stress have remained elusive. We provide evidence that the molecular basis for stress-induced mutagenesis in an *E. coli* model is error-prone DNA double-strand break repair (DSBR). I-SceI-endonuclease-induced DSBs strongly activate stress-induced mutations near the DSB, but not globally. The same proteins are required as for cells without induced DSBs: DSBR proteins, DinB-error-prone polymerase, and the RpoS starvation-stress-response regulator. Mutation is promoted by homology between cut and uncut DNA molecules, supporting a homology-mediated DSBR mechanism. DSBs also promote gene amplification. Finally, DSBs activate mutation only during stationary phase/starvation but will during exponential growth if RpoS is expressed. Our findings reveal an RpoS-controlled switch from high-fidelity to mutagenic DSBR under stress. This limits genetic instability both in time and to localized genome regions, potentially important evolutionary strategies.

## Introduction

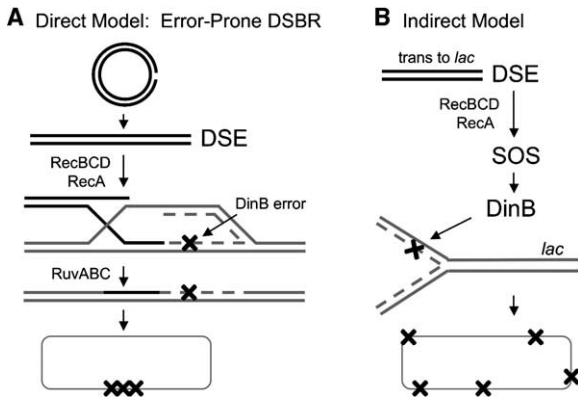
A growing body of work in bacteria and yeast has revealed the existence of special mechanisms of mutation that are induced specifically during growth-limiting stresses (reviewed in Foster [1999], Hersh et al. [2004], Rosenberg [2001], and Tenaillon et al. [2004]). These “stress-induced” or “stationary-phase” mutation mechanisms cause various kinds of genetic and genomic instability. They also occasionally produce adaptive mutations that allow growth, potentially accelerating evolution of microbial populations specifically when they are poorly adapted to their environments (Bjedov et al., 2003). Stress-induced mutation mechanisms differ from those that produce classical spontaneous mutations, which occur with a definable relationship to cell generations in proliferating cells (e.g., Lea and Coulson [1949] and Luria and Delbrück [1943]). The mechanisms underlying stress-induced genetic instability may pro-

vide improved models for the origins of genetic changes that promote aging (Finch and Goodman, 1997), evolution of resistance of pathogens to antibiotics and to host immune systems (Martinez and Baquero, 2000; McKenzie and Rosenberg, 2001), oncogenesis (Strauss, 1992), tumor progression and resistance to chemotherapy (Loeb, 1991; Yuan et al., 2000), and perhaps much of mutation-driven evolution generally. Yet, little direct evidence bears on most of these mutation mechanisms.

There appear to be multiple mechanisms of stress-induced mutagenesis observed with different stress conditions, strains, and organisms, as inferred from different genetic requirements for mutation in the different assays. However, there are some notable common components. Roles for the bacterial DNA-repair protein RecA are inferred in several, but not all, mechanisms (reviewed in Rosenberg [2001] and Rosenberg and Hastings [2003]). RecA is a widely conserved DNA-repair protein with homologs in all domains of life. In prokaryotes and eukaryotes, RecA and its homologs participate in homologous recombinational DNA repair, including DSBR, and in bacteria, RecA also functions in induction of the SOS DNA damage response (Cox, 2003). Indirect evidence suggested that an error-prone DSBR mechanism might underlie the most well-characterized, stress-induced mutation response, one of two that operate in an assay for reversion of an *E. coli lac* mutation during starvation. In this assay, cells carrying an episomal *lac* frameshift allele produce Lac<sup>+</sup> revertant colonies in response to starvation on lactose medium (Cairns and Foster, 1991; Lombardo et al., 2004; McKenzie et al., 1998). The starvation/stress-induced Lac<sup>+</sup> colonies consist of two kinds: compensatory-frameshift mutants (Lac<sup>+</sup> point mutants) (Foster and Trimarchi, 1994; Rosenberg et al., 1994) and cells carrying amplification of the leaky *lac* allele to 20–100 copies, which allow growth (*lac*-amplified clones) (Hastings et al., 2000). Though hypothesized to be an intermediate in the point-mutation pathway (Hendrickson et al., 2002), amplification is now known to be a separate stress-induced outcome and mechanism (Hastings et al. [2004] and further below). Point mutation requires RecA and other DSBR proteins, suggesting an error-prone DSBR mechanism (Foster et al., 1996; Harris et al., 1994, 1996; e.g., Figure 1A). The general strategy of coupling DSBR directly to stress-induced mutagenesis is also suggested by indirect evidence in a yeast stress-induced mutation mechanism (Heidenreich et al. [2003], discussed below). However, in both of these systems, DSBR proteins might play indirect roles in mutagenesis. For example, Lac<sup>+</sup> point mutation, but not amplification, also requires the SOS response and DinB, an error-prone DNA polymerase that it upregulates (McKenzie et al., 2000, 2001), and the DSBR proteins might act indirectly in point mutation by inducing SOS (Rosenberg, 2001; Figure 1B).

Among bacterial stress-induced mutation mechanisms, the most widespread requirement is for the bacterial general-stress-response transcriptional-activator

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**Figure 1. Models for the Mechanism of Stress-Induced Point Mutation**

Modified from Rosenberg (2001).

(A) Error-prone DSBR model. Mutations are proposed to result from DNA polymerase errors made during repair of DSEs by homologous recombination. RecBCD processes a DSE to ssDNA and loads RecA, which catalyzes strand exchange with an intact DNA copy (e.g., a sister molecule) to initiate repair. The recombination intermediate primes DNA synthesis (dashed lines), during which DinB error-prone DNA polymerase could produce errors (Xs), leading to mutations at the sites of DSBR. Possible sources of DSBs that provoke this mechanism (reviewed in [Rosenberg \[2001\]](#)) include conversion of ss nicks to DSEs upon replication. On the F', Tral endonuclease makes frequent single-strand (ss) nicks at the transfer origin, and these could become mutation-activating DSEs ([Kuzminov, 1995](#); [Rosenberg et al., 1995](#)). In the bacterial chromosome, which also displays DSBR-protein- and DinB-dependent stress-induced mutation ([Bull et al., 2001](#)), though ~20 times less than in the F' ([Foster, 1997](#)), rarer DSEs could provoke a similar mutation mechanism.

(B) Indirect model. A DSE anywhere in the cell, in *trans* to *lac*, is processed to ssDNA by RecBCD and coated by RecA. This leads to SOS induction, upregulating DinB, which creates errors at any replication fork anywhere in the genome. RuvABC resolvase could be needed to complete repair of the initiating DSB for cell survival or could be part of creating the initiating DSE by cleaving a repressed replication fork (Harris et al., 1994).

protein RpoS (e.g., Bjedov et al. [2003], Gomez-Gomez et al. [1997], Ilves et al. [2001], Lombardo et al. [2004], Rosenberg and Hastings [2003], and Saumaa et al. [2002]) or similar stress-response regulators (Sung and Yasbin, 2002). RpoS is a bacterial transcription (sigma) factor expressed in response to starvation, pH, osmotic, and temperature stresses (Hengge-Aronis, 2000). RpoS promotes transcriptional upregulation of >140 stress-response genes (Weber et al., 2005). Which RpoS-controlled function(s) is required for any stress-induced mutation mechanism has not been determined, though DinB error-prone DNA polymerase, which RpoS upregulates (Layton and Foster, 2003), is a candidate for Lac<sup>+</sup> point mutagenesis, but not *lac* amplification. Thus, though potentially critical in controlling stress-induced genetic instability, the role(s) of the RpoS stress response is not defined.

In this report, we provide direct evidence for an error-prone DSBR mechanism for stress-induced mutagenesis in the *E. coli* Lac system. We further show that RpoS controls a switch that changes the normally high-fidelity process of DSBR, via homologous recombination, to

an error-prone one under stress. This mechanism limits stress-induced genetic instability both in time, to during the stress response, and also within the genome, to regions near a DSB, and so may produce mutations that are localized (Figure 1A, bottom) rather than generally dispersed (Figure 1B, bottom). We suggest that localizing stress-induced mutagenesis could be an important evolutionary strategy, both for minimization of deleterious mutations in genomes of cells that acquire a rare adaptive mutation and also for concerted evolution within genes and gene clusters.

## Results and Discussion

## Stress-Induced Mutation and *lac* Amplification Require Tral Endonuclease

When cells carrying the F'-located *lac* +1 frameshift allele are plated onto solid minimal lactose medium, about half of the Lac<sup>+</sup> colonies visible after two days of incubation are generation-dependent revertants, formed during growth prior to plating (Cairns and Foster, 1991; Harris et al., 1999; McKenzie et al., 2003). The other half of day-two colonies (Harris et al., 1999; McKenzie et al., 2003) and those colonies from day three onward possess either stress-induced (Lombardo et al., 2004; McKenzie et al., 1998) Lac<sup>+</sup> point mutations (Foster and Trimarchi, 1994; Rosenberg et al., 1994) (as well as mutations in other, non-*lac* genes [Godoy et al., 2000; Rosche and Foster, 1999; Torkelson et al., 1997]) or *lac* amplifications (Hastings et al., 2000, 2004). The *lac*-amplified clones are infrequent initially, representing ~5% of day-five colonies, but increase to ~40% by day eight and even more later. Because DSBR-protein and DinB-error-prone-polymerase-dependent point mutation is about 20 times more frequent on the F' (Foster, 1997) than at a chromosomal site (Bull et al., 2001), we tested whether the F'-specific single-strand endonuclease Tral is required for the F'-*lac* reversions. Single strand (ss) nicks made by Tral at the origin of transfer *oriT* were hypothesized to lead to double-strand ends (DSEs) that were proposed to promote adaptive mutation in this system (Kuzminov, 1995; Rodriguez et al., 2002; Rosenberg et al., 1995). We found that Tral is required for both stress-induced point mutation and *lac* amplification (Figure 2A and Table 1). This could indicate that either ss nicks or DSEs promote mutation and *lac* amplification. We provide evidence below that DSEs are the relevant lesion.

## A Chromosomally Encoded Inducible I-SceI Endonuclease for Specific DSBs in *E. coli*

To make DSBs inducibly at specific sites in the genome, we cloned the I-SceI double-strand-endonuclease universal open reading frame (Colleaux et al., 1986) behind the *E. coli* arabinose-inducible  $P_{BAD}$  promoter and placed it in the *E. coli* chromosome (Gumbiner-Russo et al., 2001). I-SceI makes a specific DSB with 4 bp, 3' overhangs at an 18 bp cutsite that is not normally present in the *E. coli* genome (Monteilhet et al., 1990). Strains that carry both the  $P_{BAD}$ -I-SceI cassette and a cutsite either in the chromosome (data not shown) or the F' episome, but not strains carrying each singly, display 40- to 100-fold reductions in colony-forming

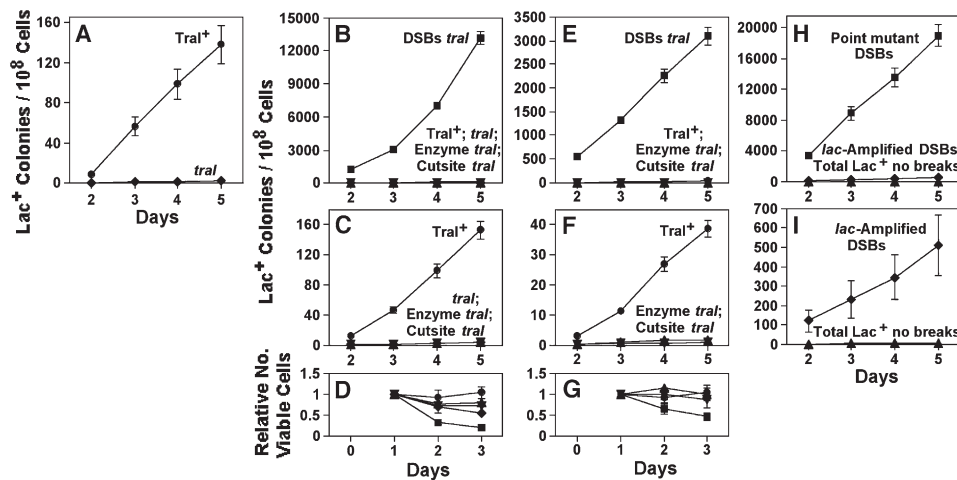


Figure 2. I-SceI Action Near *lac* Activates Adaptive Mutation, Substitutes for Tral, and Promotes *lac* Amplification

In this and Figures 3 and 4, representative examples of single experiments are shown (as cumulative plots); see Table 1 for mutation rates (per day) and their quantification from multiple experiments.

(A) Tral ss endonuclease is required for stress-induced mutation. Means  $\pm$  one SEM (error bars) for ten independent cultures per strain (A, B, and C). In this and all figures, where not visible, errors (SEM) are smaller than the symbol. Strains: Tra<sup>+</sup> (●), SMR6272;  $\Delta$ tral (◆), SMR6318. (B) I-SceI action at the *codA21* outside 11.4 kb downstream of *lac* activates mutation >1000-fold and substitutes for Tral endonuclease. Strains (B)–(D) as (A), plus: Enzyme *tral* (▲), SMR6316; Cutsite  $\Delta$ tral (▼), SMR6309; and DSBs  $\Delta$ tral (■), SMR6308. (C) Data from (B) but with the y axis expanded.

(D) Relative viability of the Lac<sup>−</sup> cells during selection (per Harris et al. [1996]). Only the DSB strains show decreased viability (2- to 5-fold) from days 1–4. The effect of viability on the appearance of Lac<sup>+</sup> mutants is offset by 2 days (Lac<sup>+</sup> point mutants take 2 days to form a colony [McKenzie et al., 1998]), so a decrease in cell number from day 2 to 3 yields a decrease in Lac<sup>+</sup> mutant number from day 4 to 5. Means  $\pm$  one SEM for four or five cultures per strain. The Lac<sup>+</sup> colony counts/cell in all figures are normalized to the number of living cells two days previously.

(E–G) As in (B)–(D), I-SceI action at *mhpA* 4.5 kb upstream of *lac* activates adaptive mutation >1000-fold. Strains: Tra<sup>+</sup> (●), SMR6272; Enzyme  $\Delta$ tral (▲), SMR6316; Cutsite  $\Delta$ tral (▼), SMR6305; and DSBs  $\Delta$ tral (■), SMR6304. Means  $\pm$  one SEM for  $n = 8$  (E and F) and  $n = 4$  (G) independent cultures per strain.

(H) Relative induction of point mutation and *lac* amplification by I-SceI. Strains (H) and (I) carrying chromosomal P<sub>BAD</sub>I-SceI  $\pm$  the F'-borne *codA21* outside follow: enzyme + cutsite (DSBs) (■, ◆), SMR6308; cutsite only (no breaks) (▲), SMR6316. Means  $\pm$  one SEM for eight independent cultures per strain. Total Lac<sup>+</sup> no breaks are all Lac<sup>+</sup> colonies scored for the SMR6316 strain. Point mutant (■) and *lac*-amplified (◆) values for the DSBs genotype were assayed based on their sectoring versus solid blue-colony phenotypes on X-gal medium, respectively (per Hastings et al. [2000]). In eight cultures from strain SMR6308,  $\leq 42$  Lac<sup>+</sup> isolates/culture/day (1334 total) were screened, and 35 of the 1334 (2.6%) were sectoring (*lac* amplified). Relative viabilities of all strains were consistent with previous observations (D and G).

(I) I-SceI action promotes *lac* amplification. Data from (H) but with the y axis expanded.

units (cfu) on arabinose medium (2%  $\pm$  0.9% survival, mean  $\pm$  SEM of data from strains SMR6304, SMR6308, SMR6310; available in the Supplemental Data with this article online) and arabinose-induced linearization of DNA (Meddows et al., 2004), indicating that most receive a DSB. Our inducible chromosomal construct has also been used for in vivo gene cloning (Zhang et al., 2002) and studies of DSBR (Meddows et al., 2004), and I-SceI expression has been a mainstay of in vivo mammalian DSBR studies (Johnson and Jasin, 2001).

#### Action of I-SceI Endonuclease Near *lac* Activates Stationary-Phase Mutation

We used the regulatable I-SceI to test whether DSBs made in the vicinity of *lac* promote mutation. Three different I-SceI cutsites were tested: one 4.5 kb upstream of the *lac* allele in the *mhpA* gene and two downstream, 10 and 11 kb away, in the *codA* gene. Cells grown with glucose, which represses P<sub>BAD</sub>I-SceI, were washed and plated onto lactose medium without additional arabinose (inducer) to allow low-level expression (Supplemental Data), and Lac<sup>+</sup> revertants were scored over

time. We found that lack of repressing glucose was sufficient to stimulate mutation and that addition of arabinose was not only not necessary for mutation but also led to greatly increased inviability (presumably due to excessive DSBs). We find that cells that carry both P<sub>BAD</sub>I-SceI and any of these cutsites near *lac* show a >1000-fold increase in Lac<sup>+</sup> mutation (Figures 2B and 2E and Table 1), and the overall cell viability decreases only about 2- to 5-fold over the course of the five-day experiments (e.g., Figures 2D and 2G). The fold increases in mutation rate (mutants per day, Table 1) are 5000- to 7000-fold for the upstream *mhpA* site and 7000- to 9000-fold and 6000- to 7000-fold for the two downstream sites, *codA22* and *codA21*, respectively. Both enzyme and cutsite are required for these increases (Figures 2B, 2C, 2E, and 2F and Table 1), implying that DSBs promoted mutation. The two downstream sites are in opposite orientations; thus, orientation of the nonpalindromic I-SceI site is not important for promoting Lac<sup>+</sup> reversion, as expected of a DSB-promoted mechanism.

21 out of 24 independent I-SceI-stimulated Lac<sup>+</sup> mu-

Table 1. Quantification of Stress-Induced Mutation Rates from Multiple Experiments

| Relevant Genotypes Compared <sup>a</sup>                         |                                                                     |                           | Temperature<br>(°C) <sup>c</sup> | Number of<br>Experiments <sup>d</sup> | Difference in Mutation Rate<br>from Control <sup>e</sup> |          |
|------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------|----------------------------------|---------------------------------------|----------------------------------------------------------|----------|
| Strain                                                           | Isogenic Control                                                    | Strain Names <sup>b</sup> |                                  |                                       | Mean                                                     | ±SEM     |
| <i>tra</i> <sup>+</sup>                                          | $\Delta traI$                                                       | SMR6272/SMR6318           | 37                               | 6                                     | 130                                                      | ±28      |
| <i>P<sub>BAD</sub>I</i> -SceI                                    | No enzyme, no<br>cutsite                                            | SMR6316/SMR6318           | 37                               | 6                                     | 1.1                                                      | ±0.07    |
| <i>codA22</i> cutsite                                            | No enzyme, no<br>cutsite                                            | SMR6311/SMR6318           | 37                               | 3                                     | 0.80                                                     | ±0.05    |
| DSBs at <i>codA22</i>                                            | <i>P<sub>BAD</sub>I</i> -SceI, no<br>cutsite                        | SMR6310/SMR6316           | 37                               | 3                                     | 7300                                                     | ±4300    |
|                                                                  | No enzyme + <i>codA22</i><br>cutsite                                | SMR6310/SMR6311           | 37                               | 3                                     | 9000                                                     | ±4500    |
| <i>codA21</i> cutsite                                            | No enzyme, no<br>cutsite                                            | SMR6309/SMR6318           | 37                               | 3                                     | 1.1                                                      | ±0.2     |
| DSBs at <i>codA21</i>                                            | <i>P<sub>BAD</sub>I</i> -SceI, no<br>cutsite                        | SMR6308/SMR6316           | 37                               | 4                                     | 6800                                                     | ±990     |
|                                                                  | No enzyme + <i>codA21</i><br>cutsite                                | SMR6308/SMR6309           | 37                               | 3                                     | 6400                                                     | ±720     |
|                                                                  | No enzyme + <i>codA21</i><br>cutsite                                | SMR6308/SMR6309           | 32                               | 3                                     | 23,000                                                   | ±5900    |
| DSBs at <i>codA21 recA</i>                                       | DSBs at <i>codA21 rec</i> <sup>+</sup>                              | SMR6509/SMR6308           | 32                               | 3                                     | 0.00085                                                  | ±0.0004  |
| DSBs at <i>codA21 recB</i>                                       | DSBs at <i>codA21 rec</i> <sup>+</sup>                              | SMR6517/SMR6308           | 32                               | 3                                     | 0.00013                                                  | ±0.00004 |
| DSBs at <i>codA21 ruvC</i>                                       | DSBs at <i>codA21 ruv</i> <sup>+</sup>                              | SMR6521/SMR6308           | 32                               | 3                                     | 0.0065                                                   | ±0.002   |
| DSBs at <i>codA21 dinB</i>                                       | DSBs at <i>codA21</i><br><i>dinB</i> <sup>+</sup>                   | SMR6755/SMR6308           | 32                               | 3                                     | 0.050                                                    | ±0.01    |
| <i>mhpA</i> cutsite                                              | No enzyme, no<br>cutsite                                            | SMR6305/SMR6318           | 37                               | 3                                     | 0.62                                                     | ±0.02    |
| DSBs at <i>mhpA</i>                                              | <i>P<sub>BAD</sub>I</i> -SceI, no<br>cutsite                        | SMR6304/SMR6316           | 37                               | 6                                     | 5200                                                     | ±1000    |
|                                                                  | No enzyme + <i>mhpA</i><br>cutsite                                  | SMR6304/SMR6305           | 37                               | 6                                     | 6700                                                     | ±2400    |
| DSBs at <i>mhpA rpoS</i>                                         | DSBs at <i>mhpA rpoS</i> <sup>+</sup>                               | SMR6780/SMR6304           | 37                               | 3                                     | 0.0068                                                   | ±0.006   |
| DSBs at <i>mhpA</i> /<br>pControl-no site                        | DSBs at <i>mhpA</i> , no<br>plasmid                                 | SMR7430/SMR6304           | 37                               | 3                                     | 0.24                                                     | ±0.2     |
| DSBs at <i>mhpA</i> /<br>pControl-no site                        | <i>trans</i> DSBs                                                   | SMR7430/SMR7419           | 37                               | 3                                     | 200                                                      | ±23      |
| <i>trans</i> DSBs                                                | <i>P<sub>BAD</sub>I</i> -SceI, pControl-<br>no site                 | SMR7419/SMR7417           | 37                               | 6                                     | 4.2                                                      | ±0.4     |
|                                                                  | No enzyme, <i>trans</i><br>plasmid site                             | SMR7419/SMR7418           | 37                               | 3                                     | 3.5                                                      | ±0.3     |
| <i>trans</i> DSBs <i>lexA</i> Ind <sup>-</sup>                   | <i>trans</i> DSBs <i>lex</i> <sup>+</sup>                           | SMR7438/SMR7419           | 37                               | 3                                     | 0.30                                                     | ±0.02    |
| <i>P<sub>BAD</sub>I</i> -SceI, pControl-<br>no site, F' homology | <i>P<sub>BAD</sub>I</i> -SceI, pControl-<br>no site, no<br>homology | SMR7432/SMR7417           | 37                               | 3                                     | 0.67                                                     | ±0.1     |
| <i>trans</i> DSBs +<br>homology                                  | <i>P<sub>BAD</sub>I</i> -SceI, pControl-<br>no site, F'<br>homology | SMR7434/SMR7432           | 37                               | 3                                     | 55                                                       | ±16      |
|                                                                  | No enzyme, <i>trans</i><br>plasmid site, F'<br>homology             | SMR7434/SMR7433           | 37                               | 3                                     | 56                                                       | ±11      |
|                                                                  | <i>trans</i> DSBs + no<br>homology                                  | SMR7434/SMR7419           | 37                               | 3                                     | 7.8                                                      | ±0.3     |
| <i>trans</i> DSBs +<br>homology <i>lexA</i> Ind <sup>-</sup>     | <i>trans</i> DSBs +<br>homology <i>lex</i> <sup>+</sup>             | SMR7439/SMR7434           | 37                               | 3                                     | 0.079                                                    | ±0.02    |

<sup>a</sup> Apart from the first line of strains, which compares the effect of *traI*<sup>+</sup> with  $\Delta traI$ , all remaining strains also carry the  $\Delta traI$  mutation. pControl is pRP9, which is identical to the I-SceI-cleavable *trans* plasmid pRP8, except that the former is uncleavable due to deletion of the central basepairs in the I-SceI site (Table 2 and Supplemental Data).

<sup>b</sup> Strains are listed as strain/isogenic control.

<sup>c</sup> Temperature at which cultures were grown prior to plating at 37° (Supplemental Data, Bacterial Growth).

<sup>d</sup> Each independent experiment is one such as those shown in Figures 2–4.

<sup>e</sup> Adaptive mutation rates (mutants per day) are the change in mutant frequency from day 4 to 5. For strains with significant loss of viability, mutant frequencies are calculated as the numbers of Lac<sup>+</sup> colonies per viable cell present on the plates two days previously (Lac<sup>+</sup> adaptive mutant cells take two days to form colonies under selection conditions [McKenzie et al., 1998]). This corrects for the slightly poorer viabilities of some strains (Figures 2–4). The mean difference in rate between each isogenic pair is an average of the differences obtained by pair-wise comparison of rates from assays run in parallel.



tants tested retain an intact I-SceI cutsite and expression cassette, as demonstrated by their retaining the arabinose-sensitivity phenotype. Sequencing revealed that the three, independent arabinose-resistant Lac<sup>+</sup> revertants all have single G to A substitution mutations in their I-SceI cutsites (Supplemental Data for the sequences). Because these cutsites were demonstrably functional when they stimulated Lac<sup>+</sup> mutation, this implies that I-SceI-induced DSBs were repaired mutagenically in these three I-SceI-induced Lac<sup>+</sup> revertants. Cutsites were also sequenced in six of the arabinose-sensitive isolates: five have the correct cutsite sequence (the sixth has an apparently silent cutsite mutation). Thus, the majority of breaks that promote mutation appear to be repaired accurately by homologous recombination, presumably from an unbroken sister DNA molecule. Moreover, growth on lactose, which is cleaved by  $\beta$ -galactosidase to produce glucose (and galactose), may have been sufficient to then repress *P<sub>BAD</sub>*-I-SceI and prevent killing in the Lac<sup>+</sup> mutant colonies, most of which retain a functional *P<sub>BAD</sub>*-I-SceI and cutsite.

I-SceI Substitutes for Tral Endonuclease

The data described above (Figures 2B–2G and Table 1) were generated in Tral-defective strains, indicating first, that the *P<sub>BAD</sub>*-I-SceI plus cutsite combinations substitute for Tral function in stationary-phase mutation, and second, that activation of mutation by I-SceI is far greater (~50 times more) than by Tral (Figures 2B and 2C and Table 1 for rates; Tral increases mutation rate 130-fold, whereas I-SceI does by an average of 6900-fold over the three sites assayed). We suggest that I-SceI-generated DSBs might be more efficient at promoting mutation than Tral-generated ss nicks, perhaps because ss nicks require a second independent event to produce a DSE that promotes mutation (Figure 1A), whereas I-SceI produces DSEs de novo. Similarly, I-SceI also activates mutation far more (~6900 $\times$ ) than ss nicks made near *lac* by the *gllp* ss nickase from bacteriophage  $\phi$ 1 (Rodriguez et al., 2002), which increased stationary-phase mutation only 10-fold. Third, these data support the idea that Tral promotes mutation by leading to DSBs, rather than F' transfer, because the latter could not occur when the F' is linearized by I-SceI but rather requires a specific ss-breakage event (Firth et al., 1996). Previous data showed that transfer is not required for stress-induced mutation in the Lac system (Foster and Trimarchi, 1995).

I-SceI-Induced Mutations Occur by Similar or the Same Mechanisms as Normal Stress-Induced Mutagenesis

The I-SceI-induced point mutagenesis might occur either by activation of a new mutation mechanism not normally operating in the starving cells or by increasing the rate of the previously active stress-induced point-mutation mechanism. Three lines of evidence indicate that DSBs simply increase the rate of the previously active mechanisms.

First, I-SceI-induced Lac<sup>+</sup> mutation requires the RecA, RecB, and RuvC homologous recombination-and-DSBR proteins and the DinB error-prone DNA polymerase

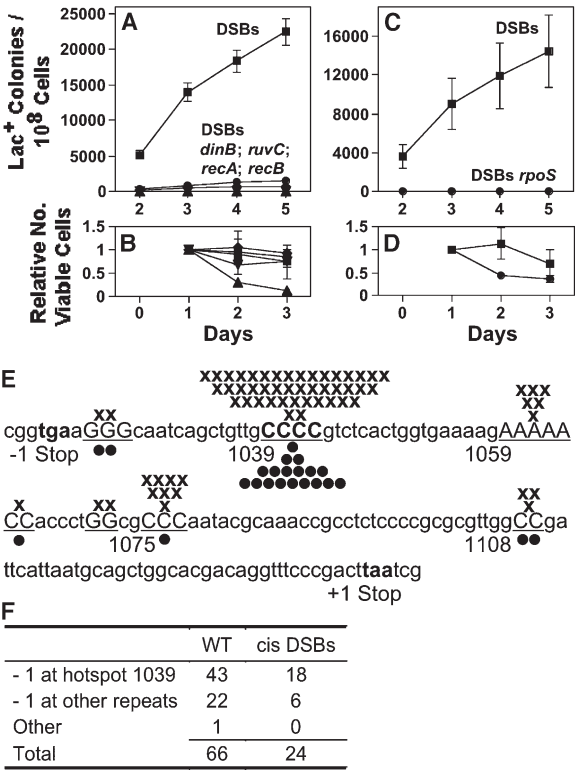


Figure 3. Requirements of I-SceI-Induced Point Mutation for DSBR Proteins, DinB, and RpoS, and Sequences of I-SceI-Induced Point Mutations (A and B) I-SceI-induced point mutation requires RecA, RecB, RuvC, and DinB. Cultures were grown at 32°C (see Supplemental Data, Bacterial Growth), which causes increased adaptive mutation in the DSB-inducing  $\Delta$ tral strain and the no-DSB-Tral<sup>+</sup> control (here and Table 1). Strains carry both *P<sub>BAD</sub>*-I-SceI, the *codA21* cutsite, and the following: Rec<sup>+</sup> DinB<sup>+</sup> (■), SMR6308;  $\Delta$ recA (▲), SMR6509; *recB* (▼), SMR6517; *ruvC* (◆), SMR6521; and *dinB* (●), SMR6755. (B) Relative viability of the Lac<sup>+</sup> cells. (C and D) I-SceI-induced mutation requires RpoS. Strains carry *P<sub>BAD</sub>*-I-SceI and the *mhpA32* cutsite: RpoS<sup>+</sup> (■), SMR6304; and *rpoS* (●), SMR6780. Means  $\pm$  one SEM for *n* = 8 (A and C) and *n* = 3 or 4 (B and D) independent cultures per strain. (E and F) Sequences of I-SceI-promoted Lac<sup>+</sup> reversions (●, *P<sub>BAD</sub>*-I-SceI and *mhpA32* cutsite) are -1 deletions in mononucleotide repeats, like normal stress-induced Lac<sup>+</sup> reversions (X, data from Foster and Trimarchi [1994] and Rosenberg et al. [1994]).

(Figures 3A and 3B and Table 1), all of which are required for stress-induced and not generation-dependent point mutation (Foster et al., 1996; Harris et al., 1994, 1996; McKenzie et al., 2001, 2003).

Second, the sequences of 24 independent I-SceI-induced Lac<sup>+</sup> point mutations are indistinguishable from the characteristic sequence spectrum of stress-induced Lac<sup>+</sup> point mutations (Figures 3E and 3F, *p* = 0.60 for rejecting the null hypothesis of similar data sets, contingency chi-square analysis). Both are mostly -1 deletions in short mononucleotide repeats, and both differ from generation-dependent Lac reversions, which are heterogeneous frameshifting mutations (Foster and Trimarchi, 1994; Rosenberg et al., 1994). 18 of the 24 I-SceI-induced point mutations occurred at the natural

hotspot, at which stress-induced Lac<sup>+</sup> reversions also focus (Figures 3E and 3F). We note that although the Lac assay selects only frameshift-reversion mutations and so does not show base substitutions, the latter should also be produced by this DinB-dependent mutation mechanism, because DinB DNA polymerase makes a specific subset of base-substitution errors as frequently as ~1 frameshift mutations (Wagner et al., 1999; Wagner and Nohmi, 2000).

Third, like normal stress-induced Lac reversions, the I-SceI-induced Lac<sup>+</sup> clones are both point mutant and *lac* amplified, and the proportions of these two distinct outcomes are similar with I-SceI induction as normally observed (following section).

All of these results imply that I-SceI does not activate a novel mutation pathway but rather increases the normal stress-induced mutation pathways already occurring, supporting the idea that DSBs are the natural intermediate for those stress-induced mutations.

In addition, because I-SceI cuts cannot substitute for DSBR proteins (RecA, RecBC, and RuvC) and DinB polymerase (Figure 3A), we infer that all of these proteins act downstream of the DSE in the mutation pathway. This is predicted by error-prone DSBR models (e.g., Figure 1A) and rules out some indirect models in which the DSBR proteins act only to create a DSE, for example via RuvC-mediated endonucleolytic cleavage of DNA at stalled regressed replication forks (Rosenberg, 2001), that then induces an SOS response and SOS-associated mutagenesis.

### I-SceI Cuts Activate Adaptive Amplification

*lac* amplification is also stimulated by I-SceI action at a site near *lac* (Figures 2H and 2I). First, introducing specific breaks near *lac* activates adaptive amplification (Figure 2I). This represents a ~260-fold stimulation in amplification rate (*lac*-amplified clones per day from days 4 to 5, calculations per Table 1 legend) over the rate of total Lac<sup>+</sup> colonies without induced DSBs (the enzyme, but-no-cutsite control). Second, the proportion of *lac* amplified among cumulative day 5 Lac<sup>+</sup> colonies is ~3% in the I-SceI-induced cells (Figure 2H and legend, *codA21* cutsite) versus ~5% in the Tra<sup>+</sup> (wild-type) control strain SMR6272 run in parallel, in which ≤24 Lac<sup>+</sup> colonies/culture/day (748 total) were screened and 37 (4.9%) were *lac* amplified (see also Hastings et al. [2000]). Experiments with the downstream *mhpA* cutsite (data not shown) gave a similar result, with ~1% of cumulative day 5 Lac<sup>+</sup> colonies due to amplification. This slightly lower frequency of *lac*-amplified clones among I-SceI-induced (1%–3%) relative to normal stress-induced Lac<sup>+</sup> colonies (5%) could be caused by I-SceI. I-SceI breaks made in the amplified DNA, after *lac* amplification, would cause extreme instability of the tandem repeats, which would then be lost. *lac* amplifications are tandem repeats usually of 7–40 kb (Hastings et al., 2000) such that many would include the nearby break sites. Also, breaks made near, but not in, an amplified array might increase instability and reduce our ability to detect some *lac*-amplified clones. Thus, I-SceI induces *lac* amplification dramatically, but it seems likely that our data underestimate the number of amplification events stimulated.

### Breaks Activate Mutation in *cis* Better than in *trans*

Error-prone DSBR models for stress-induced point mutation predict that I-SceI should activate mutagenesis preferentially in the broken DNA molecule that was repaired: in *cis* to the DSB that promoted mutation (Figure 1A). Alternatively, indirect models in which DSBs promote mutation globally in a cell (Figure 1B) predict that any DSB made anywhere in cells should activate mutation on other, unbroken DNA molecules as well: in *trans* to the mutation-activating DSB. To test whether I-SceI cuts can activate mutation in *trans*, we introduced I-SceI cuts into an unselected, multicopy (pBR322-based) plasmid, thus potentially providing many DSEs per cell, and assayed mutation in *trans* to the break, at *lac* on the F'. Our data show that our experimental conditions successfully induced plasmid cutting; we see repeatable decreases in the frequencies of ampicillin-resistant (plasmid-bearing) cells during the experiments: only 30% ± 6% (mean ± SEM, n = 6 assays) of cells carried the plasmid on day 4 on lactose medium when both P<sub>BAD</sub>I-SceI and plasmid-borne cutsite were present, whereas 95% ± 6% and 93% ± 8% did when only cutsite or enzyme was present, respectively. This decrease occurs even though cells carry about 20 copies of this plasmid. I-SceI cutting of the *trans*-plasmid site is also shown in the following section.

In contrast to I-SceI action near *lac*, I-SceI action on the pBR322-based plasmid caused only small (~4-fold) stimulations of Lac<sup>+</sup> adaptive mutation (Figure 4B and Table 1). This activation is ~200-fold less than when DSBs are provided in *cis* (Figure 4A and Table 1), despite the potentially greater number of DSEs per cell when a 20 copy-per-cell plasmid is linearized versus the one to two copy-per-cell F'. The *trans*-DSB-stimulated mutation requires an inducible SOS response (Figure 4E, open symbols), as seen by a loss of mutation in strains carrying the *lexA3*(Ind<sup>-</sup>) allele, encoding an uncleavable LexA transcriptional repressor, which prevents SOS/LexA regulon induction (Lin and Little, 1988). This suggests that although most DSB-activated mutation occurs in *cis*, as in error-prone DSBR models (Figure 1A), a small component of DSB activation occurs in *trans* via SOS induction (e.g., Figure 1B), which is required normally for stress-induced point mutation (Cairns and Foster, 1991; McKenzie et al., 2000, 2001). These conclusions are also supported by a study of the roles of the DSBR proteins and SOS response in point mutation (A.S. He, P.R. Rohatgi, M.N. Hersh, and S.M.R., unpublished data).

### Homologous Interaction Activates Mutation

A defining feature of error-prone DSBR models for point mutation is the requirement for homology between the DSE and the DNA that acquires the mutation (e.g., see Figure 1A). The homology might usually be provided by a sister DNA molecule (present in ~40% of stationary-phase cells; Akerlund et al., 1995), a duplicated genome segment, or other source of homologous DNA. We tested whether DNA ends in *trans* to *lac* (normally poorly active for mutation, see plasmid experiments above, Figure 4B) can activate mutation at *lac* if one of the ends is homologous to DNA next to *lac*. It is not

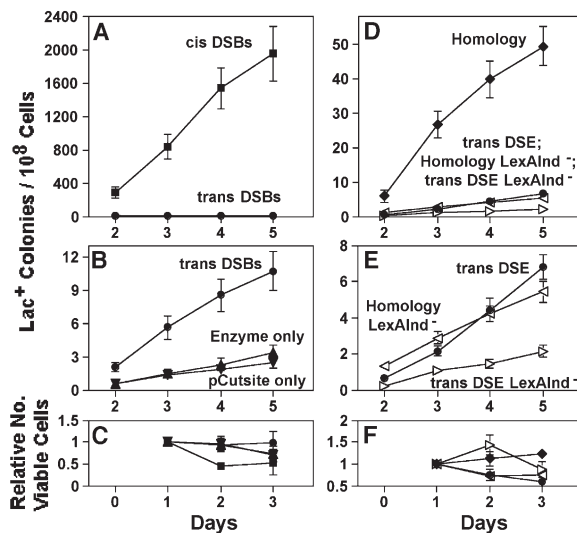


Figure 4. I-SceI Action in *cis* Activates Mutation More than in *trans*, and Homologous Interaction Activates Mutation

(A–C) I-SceI breaks in *trans* activate mutation only slightly; I-SceI breaks yield >200-fold more mutation in *cis* than in *trans* (Table 1 for rate calculations). All strains carry a Chi-containing, pBR322-derived plasmid that bears either an intact or defective I-SceI cut-site and  $\Delta tral$ , as follows: *cis* DSBs (■), SMR7430; *trans* DSBs (●), SMR7419; *P<sub>BAD</sub>*-I-SceI-no-cut-site (▲), SMR7417; and no-enzyme-pCutsite (▼), SMR7418. (B) Small effect of I-SceI action in *trans* on mutation. Data are from (A), with y axis expanded. (C) Relative viability of the Lac<sup>-</sup> cells. Means  $\pm$  one SEM for *n* = 7 or 8 (A and B) and 3 or 4 (C) independent cultures per strain.

(D) Homology between the *trans* plasmid and the F' increases mutation upon I-SceI cutting (and not without I-SceI action, Table 1). Strains (D)–(F) make *trans* DSBs with or without one of the *trans* DSEs possessing homology to 3.5 kb of DNA near *lac*, as follows: Homology (■), SMR7434; *trans*-DSEs-(no-homology) (●), SMR7419; Homology LexA(Ind<sup>-</sup>) (<), SMR7439; and *trans*-DSEs-(no-homology)-LexA(Ind<sup>-</sup>) (>), SMR7438.

(E) The small *trans* DSB stimulation of mutation requires SOS/LexA response induction, as does *trans* DSB-activated mutation in the presence of F' homology. Data from (D) but with the y axis expanded.

(F) Relative cell viabilities. Means  $\pm$  one SEM for eight (D and E) and three or four (F) independent cultures per strain

known how long the homology should be to achieve efficient creation of stable replication-competent strand-exchange intermediates in vivo (Figure 1A). However, we placed 3.5 kb of sequence identical to one of the I-SceI-cut ends of the *trans* plasmid (above) into the F' next to *lac*, oriented such that upon linearization by I-SceI, the plasmid end could interact with the F' via homology and prime DNA synthesis across *lac*. (The *lac* allele is 10 kb from where the plasmid end would pair with the F'.) I-SceI action on the *trans* plasmid in the presence of F' homology stimulated adaptive mutation ~55-fold, approximately eight to 14 times more than I-SceI action in *trans* without homology (Figure 4D and Table 1). In the no-homology case, *trans*-plasmid DSEs could still interact with other copies of the *trans* plasmid via homology, so the effect of homology on Lac<sup>+</sup> mutation is specific to interactions in the vicinity of *lac*. Both *P<sub>BAD</sub>*-I-SceI and cutsite are required for this homology-dependent activation of mutation (Table 1). As with

normal stress-induced mutation (Cairns and Foster, 1991; McKenzie et al., 2000), the homology-dependent I-SceI-activated mutation only partially requires SOS/LexA regulon induction (Figures 4D and 4E, closed symbols), implying a similar mechanism.

A potential caveat to these results is the possibility that the plasmid containing the I-SceI site integrated permanently or transiently into the F' via homology prior to cutting with I-SceI, such that the initiating DSB and mutation occurred in the same molecule. Such integration events are not expected to be frequent, but to address this possibility, 32 independent homology-stimulated Lac<sup>+</sup> isolates were screened for the presence of a cointegrated plasmid (Experimental Procedures). All 32 showed no integration in two PCR assays and a genetic assay (data not shown, Experimental Procedures). Though we cannot rule out a brief, transient integration, stable integration did not produce most DSE-dependent homology-promoted Lac<sup>+</sup> mutations.

All of these data imply that homologous interaction between a DSE and an unbroken molecule promotes mutation in the unbroken molecule. This provides strong support for error-prone DSBR mechanisms. The results also provide independent confirmation that the *trans* plasmid is acted upon by I-SceI, such that its inactivity for mutation when no homology is present (Figure 4B) cannot be attributed to failure to be cut. Our results probably underestimate the stimulatory effect of homologous ends on mutation for at least four reasons. First, because of plasmid copy numbers, each I-SceI-induced *lac*-homologous DSE would have had opportunities for ~20 homologous interactions with the *trans* plasmid itself (uncut or cut) for every one or two interactions with an F' molecule. Second, the *trans* plasmid has two ends homologous with itself and only one with the F'. Third, repair might occur preferentially with a sister molecule (not the F'). Fourth, the 3.5 kb length of homology might be shorter than would be optimal for efficient interaction with a distant, nonsister DNA molecule. These considerations might account for part or all of the ~25-fold difference in stimulation observed when comparing homologous *trans* DSEs to DSBs created in *cis* to *lac* on the F' (Table 1).

#### I-SceI Activation of Mutation Is Specific to Stationary Phase

The experiments reported above provide strong support for the existence of an error-prone DSBR mechanism of mutation in stationary-phase cells. Previous results show that the proteins required (RecA, RecBC, RuvABC, and DinB) are required for mutation only in stationary phase and not during exponential growth (Foster et al., 1996; Harris et al., 1994, 1996; McKenzie et al., 2001, 2003; Wolff et al., 2004). This could be either because DSBs are infrequent during exponential growth or because their repair is not mutagenic in log phase. We tested whether DSBs are sufficient to activate DinB-dependent mutation or whether stationary phase is also required. These experiments require that the mutation selected after log-phase growth with I-SceI induction be something other than Lac reversion, because the selection for Lac<sup>+</sup> mutants (plating on min-

Table 2. *E. coli* K-12 Strains and Plasmids Used in This Study

| Plasmid/Strain       | Relevant Genotype <sup>a</sup>                                                                                                                                                                           | Reference or Source <sup>b</sup>                         |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| pMMkatF2             | Tet <sup>S</sup> pAT153 contains <i>rpoS</i> <sup>+</sup>                                                                                                                                                | (Mulvey et al., 1988)                                    |
| pRP1                 | pMCB40 miniTn7Kan(I-SceI site)                                                                                                                                                                           | This study                                               |
| pRP3                 | pBR322 <i>tetC</i> ::miniTn7Kan(I-SceI site)                                                                                                                                                             | This study                                               |
| pRP8                 | pBR322 <i>tetC</i> ::miniTn7Kan( $\chi\chi$ I-SceI site) $\chi\chi$                                                                                                                                      | This study                                               |
| pRP9                 | pBR322 <i>tetC</i> ::miniTn7Kan( $\chi\chi\Delta$ I-SceI site) $\chi\chi$                                                                                                                                | This study                                               |
| pRP11                | pAT153 <i>tet10</i> "pControl" for pMMkatF2                                                                                                                                                              | This study                                               |
| FC29                 | $\Delta(lac-proB)_{XIII}$ <i>ara thi</i> [F' $\Delta lacI Z$ <i>proAB</i> <sup>+</sup> ]                                                                                                                 | (Cairns and Foster, 1991)                                |
| FC36                 | $\Delta(lac-proB)_{XIII}$ <i>ara thi</i> Rif <sup>R</sup>                                                                                                                                                | (Cairns and Foster, 1991)                                |
| FC40                 | FC36[F' <i>lacI33</i> $\Omega$ <i>lacZ proAB</i> <sup>+</sup> ]                                                                                                                                          | (Cairns and Foster, 1991)                                |
| SMR4562              | FC36[F' <i>lacI33</i> $\Omega$ <i>lacZ proAB</i> <sup>+</sup> ]                                                                                                                                          | Independent construction of FC40 (McKenzie et al., 2000) |
| SMR6272              | SMR4562 $\Delta araBAD567$                                                                                                                                                                               | This study                                               |
| SMR6276              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI                                                                                                                                                    | This study                                               |
| SMR6304              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>mhpA32</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                                         | This study                                               |
| SMR6305              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>mhpA32</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                                                | This study                                               |
| SMR6308              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                                         | This study                                               |
| SMR6309              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                                                | This study                                               |
| SMR6310              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>codA22</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                                         | This study                                               |
| SMR6311              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA22</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                                                | This study                                               |
| SMR6316              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' $\Delta tral$ :: <i>dhfr</i> ]                                                                                                                 | This study                                               |
| SMR6318              | SMR6272 [F' $\Delta tral$ :: <i>dhfr</i> ]                                                                                                                                                               | This study                                               |
| SMR6509              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI $\Delta(srlR-recA)306$ ::Tn10 [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                           | This study                                               |
| SMR6517              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>recB21</i> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                           | This study                                               |
| SMR6521              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>ruvC53 eda-51</i> ::Tn10 [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                             | This study                                               |
| SMR6755              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>dinB10</i> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> <i>dinB10</i> ]                                             | This study                                               |
| SMR6780              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>rpoS</i> ::Tn10 [F' <i>mhpA32</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ]                                                      | This study                                               |
| SMR7397              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1]                                                                         | This study                                               |
| SMR7402              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1 $\Delta tral$ :: <i>dhfr</i> ]                                           | This study                                               |
| SMR7403              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1 $\Delta tral$ :: <i>dhfr</i> ]                                    | This study                                               |
| SMR7407              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>dinB10</i> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1 $\Delta tral$ :: <i>dhfr</i> <i>dinB10</i> ]        | This study                                               |
| SMR7417              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' $\Delta tral$ :: <i>dhfr</i> ][pRP9]                                                                                                           | This study                                               |
| SMR7418              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' $\Delta tral$ :: <i>dhfr</i> ][pRP8]                                                                                                                  | This study                                               |
| SMR7419              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' $\Delta tral$ :: <i>dhfr</i> ][pRP8]                                                                                                           | This study                                               |
| SMR7430              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>mhpA32</i> ::miniTn7Kan(I-SceI site) $\Delta tral$ :: <i>dhfr</i> ][pRP9]                                                                   | This study                                               |
| SMR7432              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>codA23</i> ::3.5KbpRP3homology $\Delta tral$ :: <i>dhfr</i> ][pRP9]                                                                         | This study                                               |
| SMR7433              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA23</i> ::3.5KbpRP3homology $\Delta tral$ :: <i>dhfr</i> ][pRP8]                                                                                | This study                                               |
| SMR7434              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>codA23</i> ::3.5KbpRP3homology $\Delta tral$ :: <i>dhfr</i> ][pRP8]                                                                         | This study                                               |
| SMR7438 <sup>c</sup> | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>lexA3</i> (Ind <sup>-</sup> ) [F' $\Delta tral$ :: <i>dhfr</i> ] [pRP8] <sup>c</sup>                                                            | This study                                               |
| SMR7439 <sup>c</sup> | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>lexA3</i> (Ind <sup>-</sup> ) [F' <i>codA23</i> ::3.5KbpRP3homology $\Delta tral$ :: <i>dhfr</i> ][pRP8] <sup>c</sup>                           | This study                                               |
| SMR7671-SMR7694      | SMR6304 Lac <sup>+</sup> (eight each days 3, 4, and 5)                                                                                                                                                   | Independent adaptive mutants                             |
| SMR7695-SMR7726      | SMR7434 Lac <sup>+</sup> (16 each days 5 and 8)                                                                                                                                                          | Independent adaptive mutants                             |
| SMR7789-SMR7812      | SMR6304 Lac <sup>+</sup> (eight each day 3, 4, and 5)                                                                                                                                                    | Independent adaptive mutants                             |
| SMR7990              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1][pRP11]                                                                  | This study                                               |
| SMR7991              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1 $\Delta tral$ :: <i>dhfr</i> ][pRP11]                                    | This study                                               |
| SMR7992              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1 $\Delta tral$ :: <i>dhfr</i> ][pRP11]                             | This study                                               |
| SMR7993              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> I-SceI <i>dinB10</i> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1 $\Delta tral$ :: <i>dhfr</i> <i>dinB10</i> ][pRP11] | This study                                               |
| SMR7994              | SMR6272 $\Delta att\lambda$ ::P <sub>BAD</sub> [F' <i>codA21</i> ::miniTn7Kan(I-SceI site) <i>mhpC34</i> ::Tn10 <i>dtetA</i> +1][pMMkatF2]                                                               | This study                                               |

(continued)



Table 2. Continued

| Plasmid/Strain | Relevant Genotype <sup>a</sup>                                                                                                                                                | Reference or Source <sup>b</sup> |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| SMR7994        | SMR6272 $\Delta att\lambda::P_{BAD}$ [F' <i>codA21::miniTn7Kan</i> (I-SceI site) <i>mhpC34::Tn10dtetA+1</i> ][pMMkatF2]                                                       | This study                       |
| SMR7995        | SMR6272 $\Delta att\lambda::P_{BAD}$ [F' <i>codA21::miniTn7Kan</i> (I-SceI site) <i>mhpC34::Tn10dtetA+1</i> $\Delta tral::dhfr$ ][pMMkatF2]                                   | This study                       |
| SMR7996        | SMR6272 $\Delta att\lambda::P_{BAD}$ -SceI [F' <i>codA21::miniTn7Kan</i> (I-SceI site) <i>mhpC34::Tn10dtetA+1</i> $\Delta tral::dhfr$ ][pMMkatF2]                             | This study                       |
| SMR7997        | SMR6272 $\Delta att\lambda::P_{BAD}$ -SceI <i>dinB10</i> [F' <i>codA21::miniTn7Kan</i> (I-SceI site) <i>mhpC34::Tn10dtetA+1</i> $\Delta tral::dhfr$ <i>dinB10</i> ][pMMkatF2] | This study                       |
| SMR8149        | SMR6272 $\Delta att\lambda::P_{BAD}$ <i>dinB10</i> [F' <i>codA21::miniTn7Kan</i> (I-SceI site) <i>mhpC34::Tn10dtetA+1</i> <i>dinB10</i> ][pRP11]                              | This study                       |
| SMR8150        | SMR6272 $\Delta att\lambda::P_{BAD}$ <i>dinB10</i> [F' <i>codA21::miniTn7Kan</i> (I-SceI site) <i>mhpC34::Tn10dtetA+1</i> <i>dinB10</i> ][pMMkatF2]                           | This study                       |

<sup>a</sup> Most of the strains used are as SMR6272, a *ΔaraBAD567* derivative of SMR4562, but with modifications to the chromosomal and/or F' genome. For simplicity, only the additional genetic elements in each of these replicons, and additional plasmids, are indicated as follows. E.g., "SMR6272  $\Delta att\lambda::P_{BAD}$ -SceI [F' *codA21::miniTn7Kan*(I-SceI site)] [pRP11]," means that one element has been added to the chromosomal genome ( $\Delta att\lambda::P_{BAD}$ -SceI), one to the F' genome that also contains the F' genetic elements *lacI33*/*lacZ* *proAB*<sup>+</sup> present in SMR6272, and one additional plasmid, pRP11, has been added.

<sup>b</sup> The strains shown here were used in the experiments in this study. Their constructions are provided in Table S1.

<sup>c</sup> These strains also carry *malB::Tn9*.

imal lactose medium) itself induces I-SceI expression (as above) such that positive results would not be attributable unambiguously to mutation during exponential growth. We used strains carrying a *tet* +1 frameshift allele ~5 kb upstream of *lac* (Table 2; Supplemental Data), which confers tetracycline resistance (*Tet*<sup>R</sup>) when reverted, in combination with the *codA21* I-SceI cutsite downstream of *lac*. In a series of time-course experiments comparing mutant frequency with growth phase, with moderate I-SceI induction throughout (but repression upon selecting *Tet*<sup>R</sup> mutants, Supplemental Data), we found that I-SceI stimulates *DinB*-dependent reversion of the *tet* allele in stationary-phase, but not log-phase, cells (Figure 5A). I-SceI-induced mutant frequency increased 190- ± 80-fold from the log-phase level upon entry into stationary phase (mean ± SEM, Figure 5A and two additional experiments not shown). No increase was seen in the no-DSB control strain (data not shown).

Also supporting the stationary-phase specificity of I-SceI-induced *DinB*-dependent mutagenesis, I-SceI-induced mutation in the Lac assay requires the stationary-phase/general stress-response transcription factor RpoS (Figures 3C and 3D and Table 1).

These results suggest that DSBs cannot substitute for the stationary phase of growth, nor for the RpoS requirement, and thus are not sufficient to activate stress-induced point mutagenesis; one or more additional components regulate the process. The possible caveat that we failed to achieve I-SceI induction during growth, or otherwise did not provide sufficient opportunity for DSB-promoted mutagenesis, is addressed below with the demonstration that I-SceI cuts made in growing cells can promote mutation if RpoS is also expressed.

#### Amplification-Mutagenesis Model Not Supported

These results (Figure 5A) also show that stationary phase activates mutagenesis independently of lactose. A previous model for point mutagenesis postulated that strains carrying the leaky *lac* frameshift allele grow slowly in lactose medium, that *lac* amplifications are

selected for better growth in lactose, and this growth leads to mutagenesis (Hendrickson et al., 2002). The *Tet*<sup>R</sup> mutations (Figure 5A) are greatly stimulated by stationary phase without providing lactose, and thus with no selection for amplification of *lac* (or the nearby *tet* or *dinB* genes). This supports the conclusions of another recent study (Hastings et al., 2004) that the amplification-and-growth model is not tenable.

#### I-SceI and RpoS Together Activate Mutation in Exponentially Growing Cells

A possible reason for failure of I-SceI to activate mutation in log-phase cells (Figure 5A) could be that mutagenic DSBR requires one or more components produced specifically in stationary phase. The RpoS transcriptional activator upregulates >140 genes specifically in stationary phase and during some other stresses and is not normally present in exponentially growing cells (Hengge-Aronis, 2000; Weber et al., 2005). RpoS is actively removed from growing cells both by proteolytic degradation and growth-phase-specific translational inhibition; however, we attempted to achieve some level of RpoS expression during growth with a multicopy plasmid shown previously not to block cell proliferation (Farewell et al., 1998). We show that the RpoS expression plasmid in conjunction with I-SceI cutting can activate mutation during growth as follows (Figure 5B). Fluctuation tests were used to determine generation-dependent mutation rates at single midlog-phase time points (Supplemental Data). First, we see that the RpoS expression plasmid conferred a small, statistically significant ~2-fold increase in mutation rate upon wild-type (*Tra*<sup>+</sup>) cells, in the absence of additional DSBs (Figures 5B and 5C, *p* = 0.05, nonparametric Mann-Whitney rank sum test compared with vector-only control), whereas it actually decreased mutation slightly in cells with neither *Tral* nor I-SceI to make breaks. The increase in *Tra*<sup>+</sup> cells required *Tral* and *DinB* (Figures 5B and 5C), suggesting that *Tral*-generated ends provoked *DinB*-dependent error-prone repair when RpoS was expressed. Second, in the presence of I-SceI cuts, the

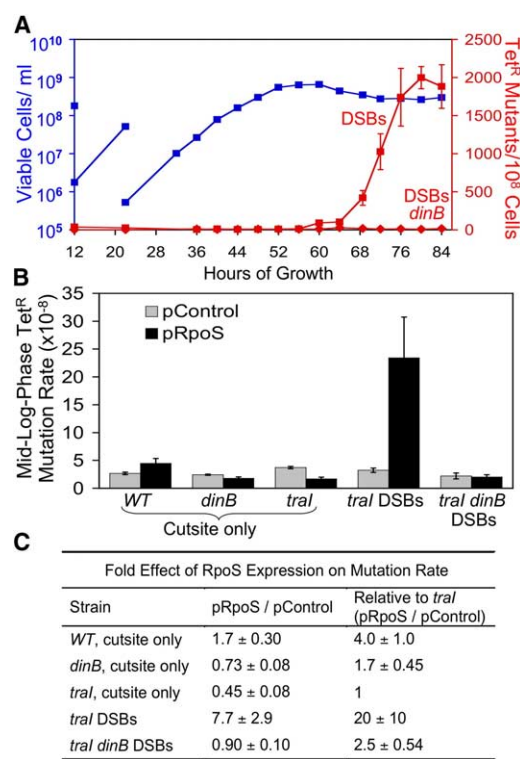


Figure 5. Either Stationary Phase or RpoS Is Required for I-SceI-Induced Point Mutation

(A) I-SceI induces mutation in stationary and not log-phase cells. Strains carry *P<sub>BAD</sub>*-I-SceI,  $\Delta$ *tral*, the *codA21* cutsite, and *tet* +1 frameshift allele near *lac*, as follows: DSBs DinB<sup>+</sup> (■), SMR7403; DSBs *dinB* (◆), SMR7407. Blue lines show the growth curve of the DSBs DinB<sup>+</sup> strain in medium with 0.000005% arabinose for I-SceI induction (Supplemental Data). Red lines show Tet<sup>R</sup> mutant frequency. Although not shown, no increase in mutant frequency was observed in log phase or stationary phase in the absence of DSBs (SMR7402, tested in parallel). Means  $\pm$  one SEM for four or five independent cultures per strain.

(B) DSBs + RpoS activate mutation in midlog phase. Mean mutation rates (mutations per cell per generation)  $\pm$  one SEM assayed from single midlog-phase time points (Supplemental Data) are averaged from *n* = 3 separate fluctuation tests for all but SMR7993, for which the mean  $\pm$  range of two experiments is shown. Strains carry either a control plasmid (pRP11, gray bars) or RpoS-expressing plasmid (pMMkatF2, black bars) and are, from left to right, SMR7990, SMR7994, SMR8149, SMR8150, SMR7991, SMR7995, SMR7992, SMR7996, SMR7993, and SMR7997. See Table 2 and Supplemental Data.

(C) Fold effect of RpoS on generation-dependent mutation in mid-log phase. pRpoS/pControl values are means  $\pm$  one SEM for the three experiments shown in (B) (two experiments for SMR7997/7993  $\pm$  range). Values normalized to the *tral* strain.

RpoS expression plasmid (in the  $\Delta$ *tral* background) increased mutation rate 20-fold (relative to its effect in *tral* cells with cutsite only, Figures 5B and 5C). This increase was also DinB dependent, not seen with the control plasmid, and required both *P<sub>BAD</sub>*-I-SceI and the cutsite to be present. Importantly, there is no evidence that RpoS expression improves the efficiency of DSBR. For example, growth rate, which is slowed by I-SceI induction in cutsite-carrying cells, is not improved by RpoS expression (Figure S1); instead, the RpoS plas-

mid slows the growth rate of I-SceI-cut cells slightly (relative to the control plasmid, Figure S1). Rather, these data imply that RpoS changes how DSBR occurs—causing a switch from high-fidelity to error-prone DSBR.

### Further Discussion

The results presented provide strong support for an error-prone DSBR mechanism for stress-induced point mutation in the Lac system and show also that amplification is provoked by DSBs. The data indicate that DNA DSBs and their repair are not always mutagenic but become so either in stationary phase (Figure 5A) or when RpoS is induced (Figures 5B and 5C), revealing the existence of an RpoS-stress-response-controlled switch from high-fidelity to mutagenic DSBR. The stationary phase/stress specificity of mutation in the Lac system was hypothesized to be conferred by an increased occurrence of DSEs in stationary-phase cells (Rosenberg, 2001). However, it now seems more plausible that spontaneous DSEs may be present continuously, given that a switch to error-prone repair occurs during stress.

RpoS might account for all or only some of that switch. Our RpoS expression experiments increased I-SceI-induced mutation rate  $\sim$ 20-fold (Figures 5B and 5C) and not the  $\sim$ 190-fold that entry into stationary phase increased mutant frequency (Figure 5A). This could be either because optimal RpoS activity was not achieved in the log-phase cells, in which multiple post-transcriptional RpoS-repression mechanisms operate (Hengge-Aronis, 2000), or because another stress response(s) or developmental program(s) during starvation might confer additional mutagenicity on DSBR (Rosenberg and Hastings, 2004). RpoS might make DSBR more mutagenic via its known 2-fold upregulation of the DinB error-prone DNA polymerase (Layton and Foster, 2005) required for stress-induced mutation in the Lac system (McKenzie et al., 2001), but there could be other components. Both RpoS (Lombardo et al., 2004) and DSBs (Figures 2H and 2I) also provoke the adaptive amplification mechanism, which is DinB independent (McKenzie et al., 2001), indicating that RpoS has at least one additional means of promoting genetic/genomic instability. RpoS was reported previously to cause mutation upon overproduction, though the relevant regulon components, and possible DSE specificity, were not examined (Yang et al., 2004).

### Evolutionary Consequences of Limiting Mutation in Time and Space

Our results show two different ways that stress-induced genetic instability is limited or controlled in the Lac system, both of which suggest important evolutionary consequences. First, the RpoS response limits genetic instability to times of stress. One does not know whether stress-induced mutagenesis, via any mechanism, evolved by selection for mutagenesis per se or whether it is an unavoidable byproduct of other stress-induced processes. However, increasing mutation rate specifically during stress might confer a selective advantage (Bjedov et al., 2003). Although most mutations are deleterious, failure to change genetically when

poorly adapted to an environment could be fatal. In large cell populations, promotion of rare adaptive mutations could be advantageous for a cell clone, even though many individual cells would be harmed by inducing mutagenesis. Coupling of mutagenesis to stress responses restricts the risk to desperate times. Control of mutation by known stress responses is common to many different stress-induced mutation mechanisms (reviewed, [Introduction](#); [Bjedov et al., 2003](#); [Lombardo et al., 2004](#)). We suggest that these might all be evolved mechanisms that restrict mutation to times of poor adaptation (stress).

Second, coupling of mutagenesis to DSB repair limits mutations in the genome, at least to the DNA molecules undergoing repair ([Figure 4](#)) and possibly to localized regions within a repaired molecule, as depicted in [Figure 1A](#), bottom. Different kinds of DSB repair reactions predict shorter or longer tracts of new DNA synthesis ([Haber et al., 2004](#)). The short-tract reactions (DSBR) should localize mutations. But possibly long-tract repair (DSE repair) might too if error-prone DNA polymerases stayed relatively briefly with replication forks initiating repair. If mutations are localized, one might expect that all or most genomic regions might be mutated in a population (as observed in [Godoy et al. \[2000\]](#), [Rosche and Foster \[1999\]](#), and [Torkelson et al. \[1997\]](#)), but only small segments in any given cell, if spontaneous DSEs are relatively infrequent per cell and randomly distributed in genomes. Localizing mutagenesis ([Figure 1A](#), bottom) could reduce deleterious mutations that would otherwise accumulate in the same cells with rare adaptive mutations, and so decrease potential benefits of stress-induced mutagenesis, and also might facilitate concerted evolution within genes and gene clusters. Although we measured frameshift mutations, DinB also produces frequent base substitutions ([Wagner et al., 1999](#); [Wagner and Nohmi, 2000](#)), such that substitutions, which can modify protein and RNA functions, are also expected. Evolution of new functions of RNAs, proteins, and protein machines often requires more than one mutation in a relatively small genomic area (a gene or gene cluster) ([Camps et al., 2004](#)), which might be facilitated by localization of stress-induced mutagenesis. This strategy is used in immune diversification, in which mutations clustered near DNA breaks produce altered immunoglobulin genes, and the mutations are made by error-prone DNA polymerases similar and homologous to DinB ([Neuberger et al., 2003](#)). Even if long tracts of repair replication are mutagenic throughout, genomes would not be mutagenized evenly; in DSE-repair reactions in which replication forks progress to the end of the genome, they do so unidirectionally toward the terminus ([Kuzminov, 1995](#)). Thus, terminus-proximal genes would be traversed by repair-replication forks more frequently, and so would be more heavily mutagenized. Genomes might have evolved with genes in which more mutations would be tolerated/useful near the terminus.

Supporting the idea of its potential utility, the basic strategy of coupling stress-induced mutagenesis to DSB repair appears to have evolved more than once. A stress-induced mutation mechanism in budding yeast

uses proteins required for nonhomologous end joining (NHEJ) ([Heidenreich et al., 2003](#)). NHEJ is a fundamentally different mechanism of DSB repair from homologous recombinational repair (discussed here), implying an independent origin of this mutation mechanism. Also in yeast, DSB repair by homologous recombination is mutagenic in rapidly growing cells ([Holbeck and Strathern, 1997](#)). It is unknown whether this might increase during stress, as shown here. Finally, although several bacterial stress-induced mutation mechanisms require RecA, not all of them do so for promotion of the SOS response, suggesting possible involvements of recombinational repair ([Rosenberg, 2001](#); [Rosenberg and Hastings, 2003](#)). It seems likely that both limitation of genetic instability in time, by coupling mutagenesis stress responses, and to local genomic regions, by coupling to processes such as repair, may emerge as general themes in independently evolved stress-induced mutagenesis mechanisms that may promote evolution of microbes.

In vertebrates, the process of somatic hypermutation of immunoglobulin genes ([Neuberger et al., 2003](#)) may be evolutionarily related to the DSB-coupled, stress-induced mutagenesis described here. Both are highly regulated and limited to specific developmental stages (pre-B cells and stress, respectively), and both use pol Y family error-prone DNA polymerases and appear to focus mutations near breaks ([Diaz and Lawrence, 2005](#)). The mammalian ortholog of bacterial DinB, the error-prone DINB1 polymerase, is normally expressed in few mammalian tissues ([Velasco-Miguel et al., 2003](#)) but is overproduced significantly in nonsmall-cell lung cancers in which it promotes tumorigenesis ([Bavoux et al., 2005](#); [Wang et al., 2001, 2004](#)). The mechanisms of genetic instability promoting these and other cancers may be similar to mutation mechanisms such as those described here. Regulated, inducible mutation mechanisms might be conserved across evolution, perhaps for functions like immune diversification or others. These mechanisms could promote cancer when the important regulatory controls, such as those described here, that limit the mutagenesis to specific cell types and genomic regions are lost.

## Experimental Procedures

### Bacterial Strains, Plasmids, Growth, and Stress-Induced Mutation Assays

*E. coli* strains, plasmids, and new genetic elements used ([Table 2](#)) were constructed as described ([Supplemental Data](#)). See the [Supplemental Data](#) for growth conditions. Strains with both  $P_{BAD}$ -SceI and DSB plasmid pRP8 show  $\leq 10$ -fold decrease in ampicillin-resistant, but not total, cfu on M9 glycerol 0.1% arabinose. However, *lexA3*(Ind<sup>-</sup>) derivatives of plasmid-DSB strains show a  $\sim 100$ -fold decrease in total cfu on M9 glycerol 0.001% arabinose.

Stress-induced mutation assays were as described ([Harris et al., 1996](#)), with modifications described in the [Supplemental Data](#). The 24 DSB-activated Lac<sup>+</sup> point mutants assayed for arabinose sensitivity (SMR7789–7812) were purified on M9 B1 glucose fucose rifampicin X-gal plates (to repress  $P_{BAD}$ , exclude FC29 scavenger cells, and identify and exclude *lac*-amplified clones), grown in M9 B1 glycerol fucose to saturation, washed twice, and cfu assayed on M9 B1 glucose and M9 B1 glycerol arabinose (0.001%). The 24 Lac<sup>+</sup> mutants sequenced (SMR7671–7694) were purified on M9 B1 lactose plates and sequenced as previously ([McKenzie et al., 2001](#)).

The 32 *trans*-plasmid-plus-homology Lac<sup>+</sup> isolates screened for the presence of a cointegrated plasmid (SMR7695–7726) were puri-



fied on M9 B1 lactose plates, grown to saturation in M9 B1 glucose, screened for arabinose resistance and ampicillin sensitivity, and tested for positive PCR by using primers spanning the homology insertion site (showing no plasmid integration) and negative PCR with a plasmid/*codA* set of primers.

### Generation-Dependent Mutation Experiments

For assays of Tet<sup>R</sup> mutant frequencies in time-course experiments (e.g., Figure 5A), five cultures per strain were inoculated from single colonies on glucose plates into M9 B1 glycerol with 0.000005% arabinose (for I-SceI induction) and grown and assayed for tetracycline-resistant (Tet<sup>R</sup>) mutant and total cfu as described (Supplemental Data), and the cultures were tested for 0.001% arabinose sensitivity at the final time point. Experiments in which a single midlog-phase time point was assayed for Tet<sup>R</sup> mutation (e.g., Figures 5C and 5D), growth curves, and mutation assays were performed as described (Supplemental Data).

### Supplemental Data

Supplemental data include Supplemental Results, Supplemental Experimental Procedures, one figure, and one table and can be found with this article online at <http://www.molecule.org/cgi/content/full/19/6/791/DC1/>.

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